

Computer Networks Fundamentals

Unit 1: Introduction to Computer Networking

Introduction to Computer Networking

A computer network is an interconnected collection of autonomous computing devices that are capable of exchanging information and sharing resources with one another. The devices that make up a network, often referred to as nodes or hosts, are connected through a combination of physical transmission media, such as copper cable and optical fibre, and wireless transmission media, such as radio waves. The study of computer networking is concerned with the rules, hardware and software that allow this exchange of information to take place reliably, efficiently and securely. At the heart of every network lies the concept of a protocol, which is a formally agreed set of rules that governs the format, timing and sequencing of the messages that are exchanged between two communicating devices. Without a shared protocol, two devices manufactured by different vendors would have no common basis on which to interpret the electrical or optical signals passing between them, and the modern era of open, interoperable networking would not be possible.

Goals of Networking and Communication Models

The principal goals that motivate the design of a computer network include resource sharing, which allows expensive hardware such as printers and storage arrays to be used by many people; reliability, which is achieved by providing alternative communication paths so that the failure of one link does not isolate an entire organisation; cost reduction, achieved by allowing many inexpensive personal computers to share the resources of a smaller number of powerful servers; and scalable communication, which allows electronic mail, video conferencing and file transfer to take place instantly between people who may be physically located on opposite sides of the world. To reason about how communication actually takes place, network designers rely on layered reference models, of which the seven-layer Open Systems Interconnection model and the four-layer TCP/IP model are the most widely taught. Each layer in such a model is responsible for a specific aspect of communication, ranging from the physical transmission of raw bits, through the reliable delivery of data segments, up to the presentation of information in a form that a human-facing application can use, and every layer communicates with its counterpart on the receiving machine through the protocol that has been defined for that layer.

Types of Networks and Network Services

Networks are commonly classified according to their geographic scope. A Local Area Network, or LAN, connects devices within a single building or campus and typically offers high transmission speed over a comparatively short distance. A Metropolitan Area Network spans a larger area such as a city, while a Wide Area Network, or WAN, connects devices that may be separated by entire countries or continents, often by leasing transmission capacity from telecommunications carriers. In addition to this classification by scale, networks are described by the services they offer to the applications that run on top of them. A connection-oriented service establishes a dedicated logical path before any data is exchanged and guarantees that packets will arrive in the order in which they were sent, whereas a connectionless service transmits each packet independently, with no guarantee of ordering, and relies on higher layers to detect and correct any resulting problems. The choice between these two styles of service has a direct influence on the design of the protocols, such as the Transmission Control Protocol, that applications use to communicate reliably across an unreliable underlying network.

Applications of Computer Networks

The practical applications of computer networking now touch nearly every aspect of daily and professional life. Electronic mail, instant messaging and video conferencing allow people to communicate instantly regardless of physical distance. The World Wide Web, which is built on top of network infrastructure using the hypertext transfer protocol, gives billions of people access to information, commerce and entertainment. Cloud computing platforms rely

on high-capacity data centre networks to deliver storage and processing power to businesses that would otherwise need to purchase and maintain their own hardware. Wireless networking technologies extend this connectivity to mobile devices, allowing a smartphone or a laptop computer to remain connected to the internet while its user moves from one location to another. Every one of these applications ultimately depends on the disciplined, layered exchange of protocol data units, and on the underlying processes of packet formation, routing and error control that this unit introduces as the foundation for the remainder of the course.

Prepared for classroom distribution as part of the Acadrium academic content repository. For internal educational use only.